Lab 6 Worksheet — IPv4 Subnetting and Address Planning with IP Calculator

Name: ____ **Date:** __

Exam objective: Calculate subnet masks, network and broadcast addresses, usable host ranges and CIDR notation, and design an address plan for a SOHO network (Core 1 objective 2.5).

Record as you go

Step	What you did	What you observed
1	Open IP Calculator at https://alfredang.github.io/ipcalculator/ and select the IPv4 tab.	
2	Enter 192.168.10.75/24 and record the network address, broadcast address, first and last usable host, netmask in dotted decimal, and the usable host count.	
3	Change the prefix to /26 with the same address and record how the network address, broadcast address and usable host count change.	
4	Verify the /26 result by hand: 32 minus 26 leaves 6 host bits, 2 to the power 6 is 64 addresses per block, minus 2 reserved gives 62 usable hosts. Confirm this matches the tool.	
5	Enter 172.16.0.0/12 and 10.0.0.0/8 in turn, and confirm from the output that all three RFC 1918 private ranges are non-routable on the internet.	
6	Enter 169.254.14.9/16 and record what this range signifies — an APIPA address assigned when no DHCP server answered.	
7	Design a SOHO plan from the 192.168.20.0/24 block for four departments needing 50, 25, 10 and 5 hosts. Choose the smallest prefix that	

Step	What you did	What you observed
	satisfies each requirement.	
8	Verify each chosen prefix in IP Calculator: /26 gives 62 usable for the 50-host subnet, /27 gives 30 for the 25-host subnet, /28 gives 14 for the 10-host subnet and /29 gives 6 for the 5-host subnet.	
9	Switch to the Batch tab, paste all four subnets one per line with a comment naming each department, and process them together.	
10	Export the batch results to CSV using the export control, and keep the file as the address-plan deliverable.	
11	Open the Killercoda Ubuntu playground and install ipcalc to cross-check your work with an independent tool.	
12	Cross-check the first subnet on the command line and confirm the network, broadcast and host range match IP Calculator exactly.	

Verification

Success criterion: Every subnet in your plan satisfies its host requirement with no wasted block larger than necessary, the hand calculation matches the IP Calculator output for the /26 case, ipcalc agrees with the browser tool, and the CSV export contains all four subnets.

- I completed every step in the lab.
- My result meets the success criterion above.
- I recorded my evidence (screenshots, output, completed tables).

Reflection

What surprised you in this lab?

Where would you apply this on the job?

What do you still need to revise before the exam?